Recap and Probability Rules

Course information

- **We have weekly homework which will be released every Friday and due on next Friday**
  - Homework this week will cover three lectures (Sep 24, 26, and 28)
- **We have two midterms and a final exam; the dates are to be determined.**
- **Final grade formular:**

A	Homework
B	In-class activity and class participation
C	Midterm exam I
D	Midterm exam II
E	Final exam

max{(.15A+.15B + .2C + .2D + .3E), (.15A+.15B + .3C + .1D + .3E), (.15A+.15B + .1C + .3D + .3E)}

Course information

- **Course materials**
  - http://courses.engr.illinois.edu/ece313/fa2020/probabilityJan25.pdf
  - Sheldon M. Ross, Introduction to Probability Models, 11th edition, Academic Press, 2014 (available online)

- **Teaching assistant**

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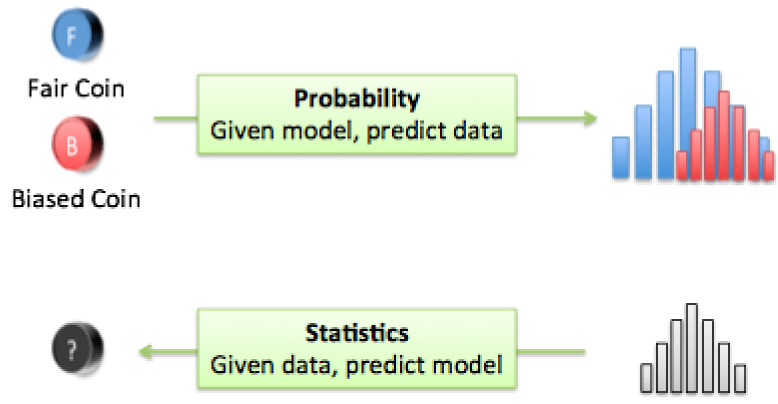
- **Office hours: by appointment (send us email)**

Difference between Probability and Statistics



Statistics:

- Summary of the data
- Estimate parameters
- Test hypotheses

Probability:

- Probabilities of events
- Expected values
- Variance

Question 1

A computer system contains 50 executable files and is attacked by a virus.

Problem 1: During the attack, each file is affected with unknown probability p. We observe that 15 files are affected. Estimate p.

Problem 2: During the attack, each file is affected with probability 0.2. How likely is it that at least 15 files get affected?

A. Problem 1 is a probabilistic problem and Problem 2 a statistical problem.

B. Problem 1 is a statistical problem and Problem 2 a probabilistic problem.

C. Both are probabilistic problems.

D. Both are statistical problems.

Three Components in Probability

- Sample space
- Event
- Probability function

Sample spaces

Definition:
An experiment is an activity or procedure that produces distinct possibilities, called outcomes.

In probability and statistics, randomness is involved: outcome of experiment is uncertain.

The set of all outcomes is called the sample space.

Notation:
Sample space: $\Omega = \{\dots\}$.
Outcome a is an element of the sample space: $a \in \Omega$.

Examples

Experiment I: Throw a coin. Sample space?

$\Omega = \{\text{Heads, Tails}\}$

Experiment II: Throw one die. Sample space?

$\Omega = \{1, 2, 3, 4, 5, 6\}$

Experiment III: Waiting time for next bus. Sample space?

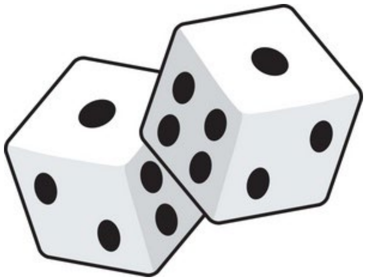
$\Omega = [0, \infty)$

Experiment IV: Output of 4 calls of random number generator (0 or 1). Sample space?

$$\begin{aligned} \Omega &= \{(0,0,0,0), (0,0,0,1), \ldots, (0,1,1,1), (1,1,1,1)\} \\ &= \{(x_1, x_2, x_3, x_4) : x_i \in \{0,1\}\} \end{aligned}$$

Question 2

We throw one die twice. The set of possible outcomes of this experiment is given by



A. $\Omega = \{2, 3, \ldots, 12\}$

B. $\Omega = \{(1,1), (1,2), \ldots, (6,5), (6,6)\}$
$= \{(i,j) \mid 1 \leq i, j \leq 6\}$

C. $\Omega = \{(1,1), (1,2), \ldots, (5,6), (6,6)\}$
$= \{(i,j) \mid 1 \leq i \leq j \leq 6\}$

D. $\Omega = \{\{1,1\}, \{1,2\}, \ldots, \{5,6\}, \{6,6\}\}$

Events

Definition:
An <u>event</u> is a subset of the sample space.

Examples:
Throw a coin three times: “Heads is thrown once” is an event.
Throw a die twice: “Sum is eight” is also an event.

Probability function

Definition:
Let Ω be a sample space. A probability function P assigns to each event A in Ω a number $P(A)$ in $[0, 1]$ such that:

1. $P(\Omega) = 1$.
2. $P(A \cup B) = P(A) + P(B)$, if A and B are disjoint.

The number $P(A)$ is called the probability of A.

Example

Experiment: month of birth of random person.
Sample space: $\Omega = \{\text{Jan, Feb, \ldots, Nov, Dec}\}$.

~~Assume equal probabilities for all months.~~

Can you think of another relevant probability function?

Let A be the event $A = \{\text{Born in an even month}\}$.

Does $P(A)$ change?

Sum and complement rule

Theorem:
For any two events A and B we have:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Theorem:
For any event A we have:

$$P(A^c) = 1 - P(A).$$

Conditional probability

Definition:
The <u>conditional probability</u> of A given C is defined as:

$$P(A \mid C) = \frac{P(A \cap C)}{P(C)},$$

provided $P(C) > 0$.

Interpretation:
Restrict the sample space to C. Then the conditional probability of A given C is the probability of A in the reduced sample space C.

Question 3

Meet Steven. Steven is a real outdoorsman. He enjoys walks trough the forest and is very fond of animals. Steven is physically very fit.

Is it more probable that Steven is a teacher or a park ranger?

A. Teacher

B. Park ranger

Multiplication rule

A box contains 10 pebbles of which 3 are granite.
You take out 2 pebbles (without looking).
What is the probability that both are granite?

Theorem:
For any two events A and C it holds that:

$$P(A \cap C) = P(A \mid C)P(C).$$

Question 4

Consider the following two statements:

1. $P(A \cap B) = P(A) + P(B) - P(A \cup B)$
2. $P(A \cap B) = P(A)P(B)$

Which of these statements is/are true?

A. None

B. Only 1

C. Only 2

D. Both

Question 5

Consider the following two statements:

1. $P(A \mid C) = P(C \mid A)$
2. $P(A \cap C) = P(C \mid A)P(A)$

Which of these statements is/are true?

A. None

B. Only 1

C. Only 2

D. Both

Question 6

Which of the following statements is true?

1. $P(A) = P(A \mid C) + P(A \mid C^c)$
2. $\dfrac{P(C \mid A)}{P(A \mid C)} = \dfrac{P(C)}{P(A)}$

Which of these statements is/are true?

A. None

B. Only 1

C. Only 2

D. Both

Law of total probability

Theorem:
Let A and C be two events. We have:

$$P(A) = P(A \mid C)P(C) + P(A \mid C^c)P(C^c).$$

Theorem:
Suppose $C_1, C_2, \ldots, C_m$ are disjoint events such that

$$C_1 \cup C_2 \cup \cdots \cup C_m = \Omega.$$

For any event A we have:

$$P(A) = P(A \mid C_1)P(C_1) + P(A \mid C_2)P(C_2) + \ldots + P(A \mid C_m)P(C_m).$$

Bayes' Rule

Theorem:
Suppose the events $C_1, C_2, \ldots, C_m$ are disjoint and fill up the sample space Ω. Then the conditional probability of C_i given some event A is given by

$$\mathrm{P}(C_i \mid A) = \frac{\mathrm{P}(A \mid C_i)\mathrm{P}(C_i)}{\mathrm{P}(A \mid C_1)\mathrm{P}(C_1) + \ldots + \mathrm{P}(A \mid C_m)\mathrm{P}(C_m)}$$

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$$P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$$

- This leads to the following **usual definition of independence**:

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- Note: If A and B are not independent, then $P(A \cap B)$ is computed using the multiplication rule.

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Check: since there are 10 x15 ways of choosing both defective chips, and there are 100 x 300 ways of choosing any two chips. Thus:

$$P(A \cap B) = \frac{10 \cdot 15}{100 \cdot 300} = 0.005 = 0.10 \cdot 0.05 = P(A)P(B).$$

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- If A and B are two mutually exclusive events, then $A \cap B = \varnothing$, (Null Set) which implies $P(A \cap B) = 0.$ Now, if they are independent as well, then either P(A)=0 or P(B)=0 (or both).

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- Note: If the events A and B are independent, and the events B and C are independent, then events A and C need not be independent (i.e., independence is not a transitive relation).

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- Are ‘red card’ and ‘ace’ independent?
- Are they Mutually exclusive?

Some Important Points about the Concept of Independence (cont.)

- If the events A and B are independent, then so are events $\overline{A}$ and B, events A and $\overline{B}$, and events $\overline{A}$ and $\overline{B}$.

Example 1: Demonstrating Dependence

- Suppose that we toss 2 fair dice. Let E_1 denote the event that the sum of the dice is 6 and F denote the event that the first die equals 4. Are these two events independent?

Example 1 continued

- Now suppose that we let E_2 be the event that the sum of the dice equals 7. Is E_2 independent of F ?

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- It follows that an appropriate definition of the independence of three events, E, F, and G would have to go further than merely assuming that all of the pairs of events are independent. We are thus led to the following definition.

Independence of E,F,G

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- Three events, E, F, and G are said to be independent if

$$P(EFG) = P(E)P(F)P(G)$$
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$$\begin{aligned} P[E(F \cup G)] &= P(EF \cup EG) \\ &= P(EF) + P(EG) - P(EFG) \\ &= P(E)P(F) + P(E)P(G) - P(E)P(FG) \\ &= P(E)[P(F) + P(G) - P(FG)] \\ &= P(E)P(F \cup G) \end{aligned}$$

Independence of n Events

- We can also extend the definition of independence to more than three events. The events $E_1, E_2, ..., E_n$ are independent if for every subset $E_{1'}, E_{2'}, ..., E_{r'}, r \leq n$ of these events,

$$P(E_{1'}, E_{2'}, ..., E_{r'}) = P(E_{1'})P(E_{2'})...P(E_{r'})$$

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- Finally, we define an infinite set of events to be independent if every finite subset of those events is independent.

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Two independent tosses of a coin.
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If the events A1, A2, …, An, are such that every pair is independent, then they are called **pairwise independent**. It does not follow that the list of events is **mutually independent**.

Another Example: Pairwise and Mutually independent

- Consider the experiment of tossing two dice. Let the sample space $S = \{(i,j) | 1 \le i,j \le 6\}$. Also assume that each sample point is assigned a probability 1/36. Define the events A, B, and C so that:

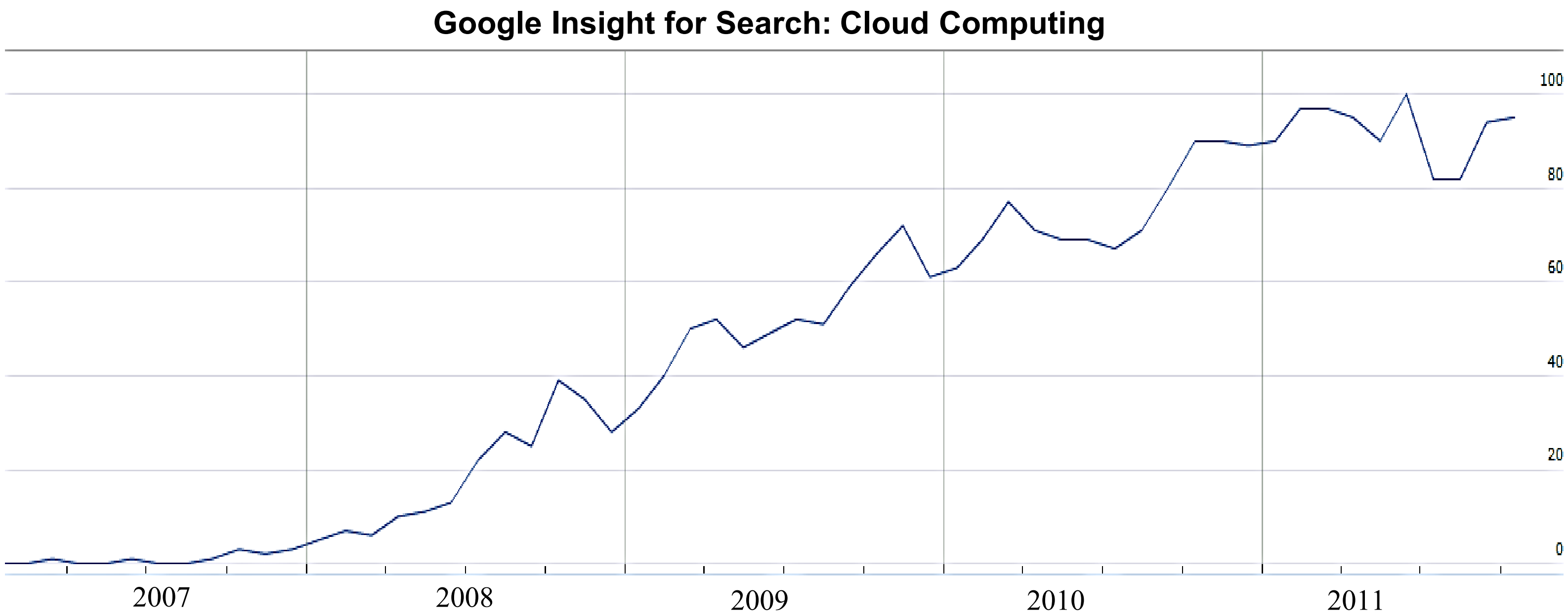
A = "First die results in a 1, 2, or 3."

B = "First die results in a 3, 4, or 5."

C = "The sum of two faces is 9."

Reliability Evaluation Applications

Calculating the Probability of failure: Cloud Computing Example



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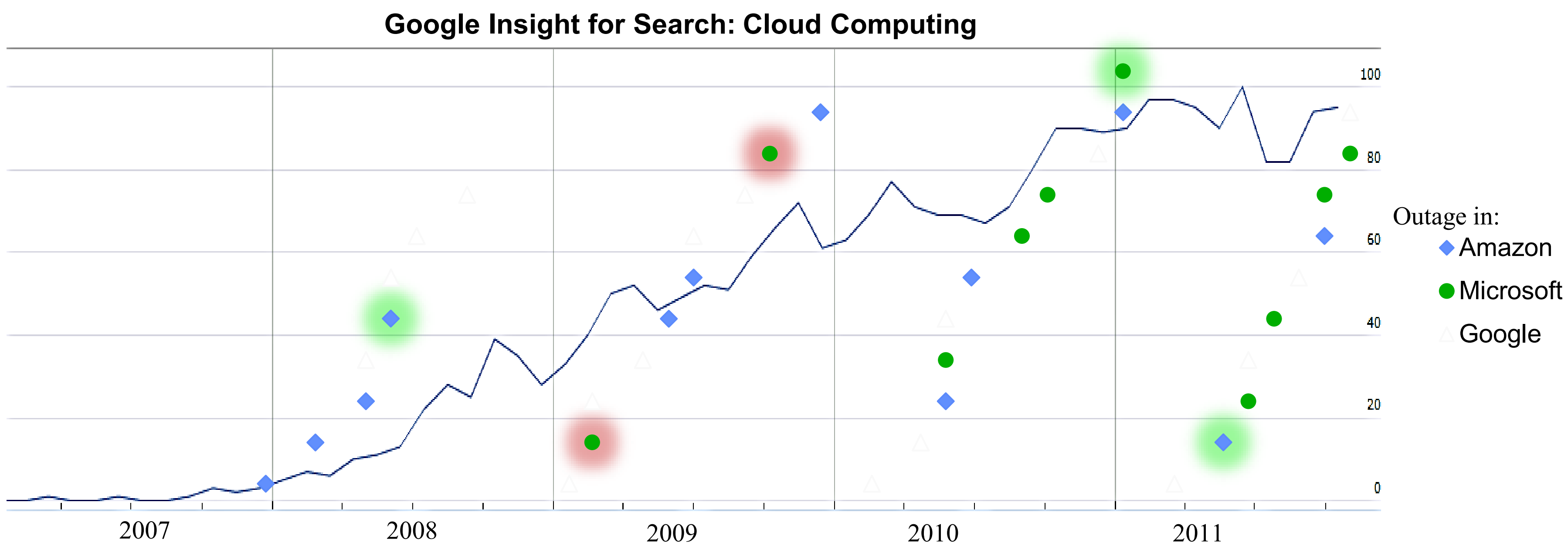

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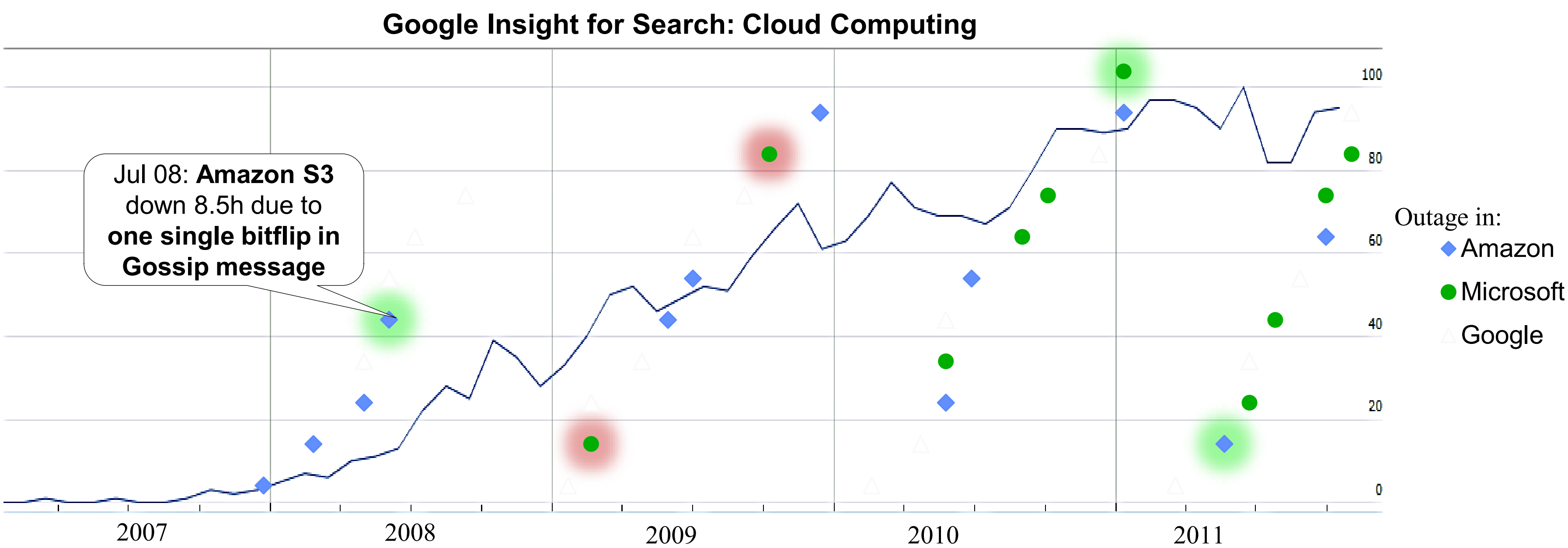

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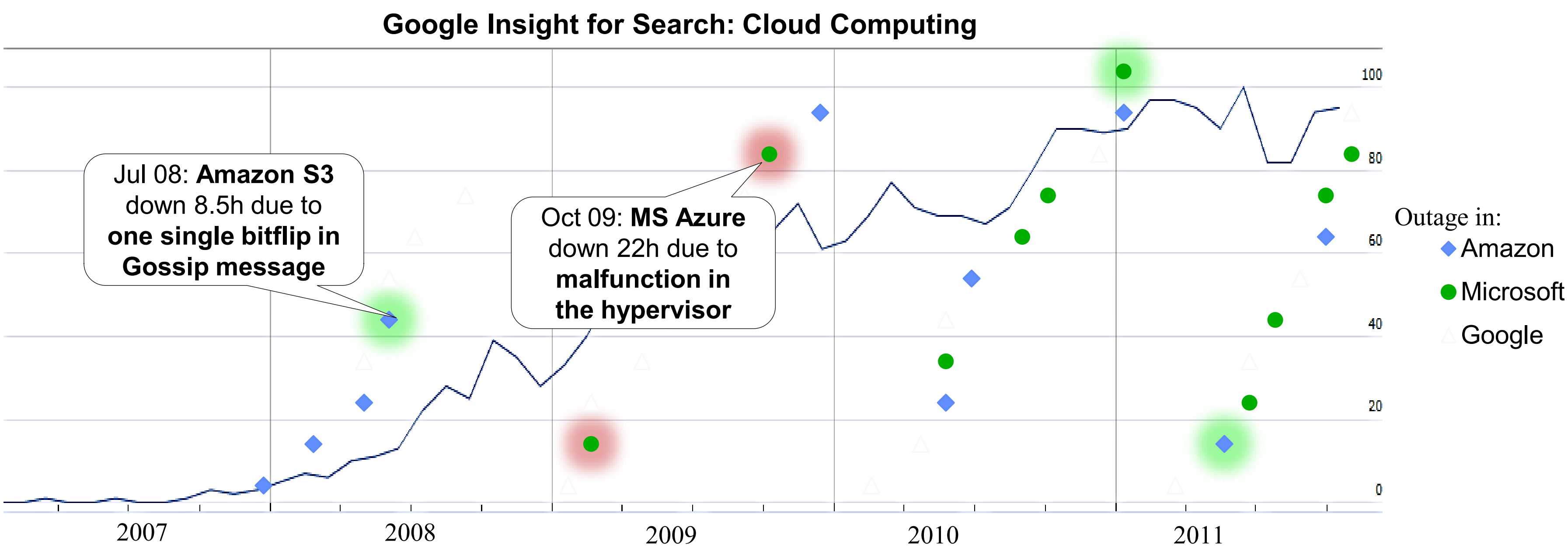

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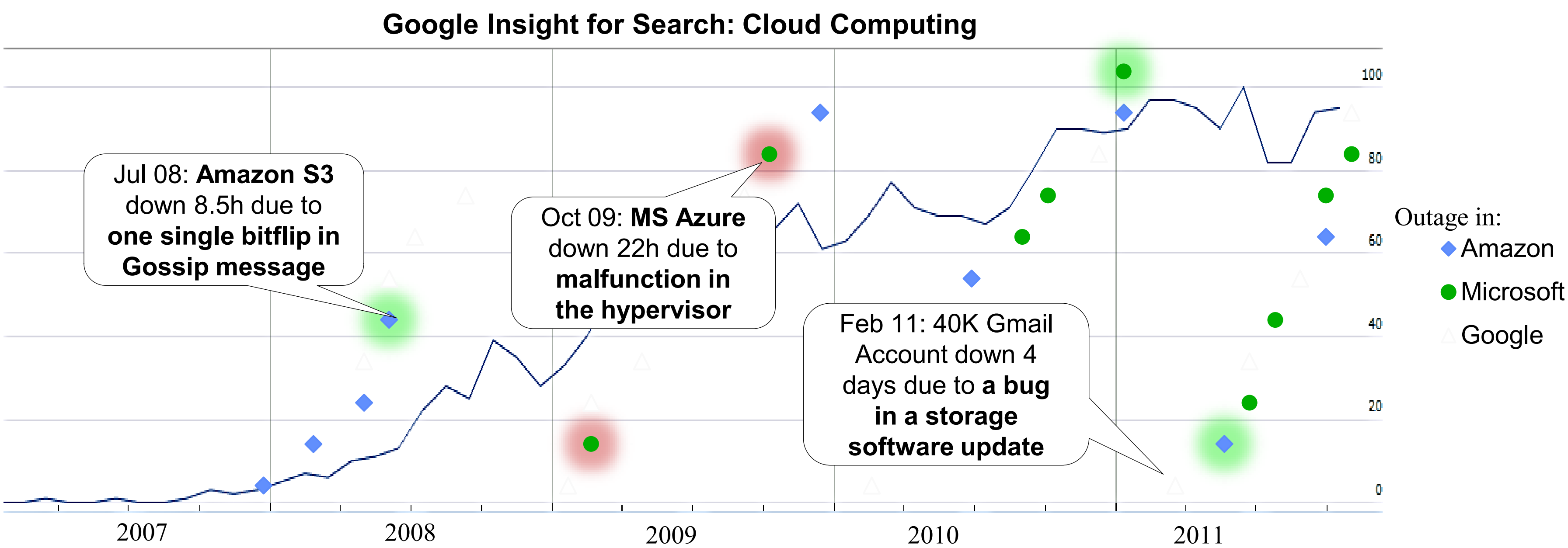

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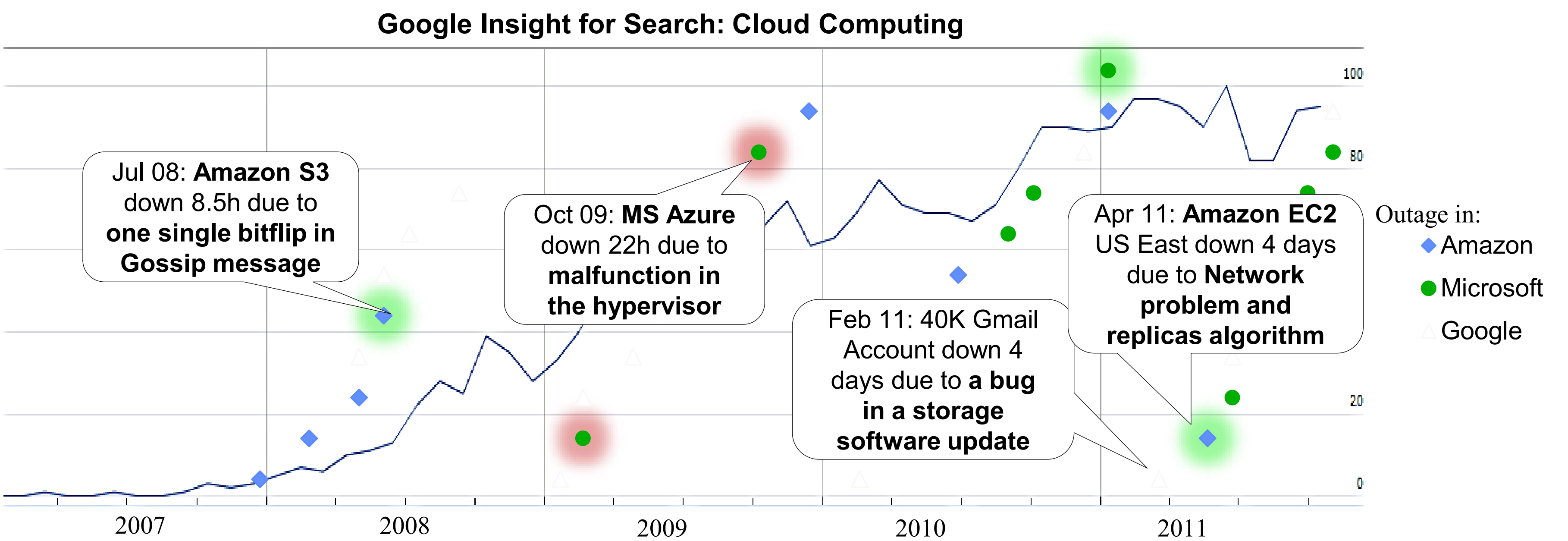

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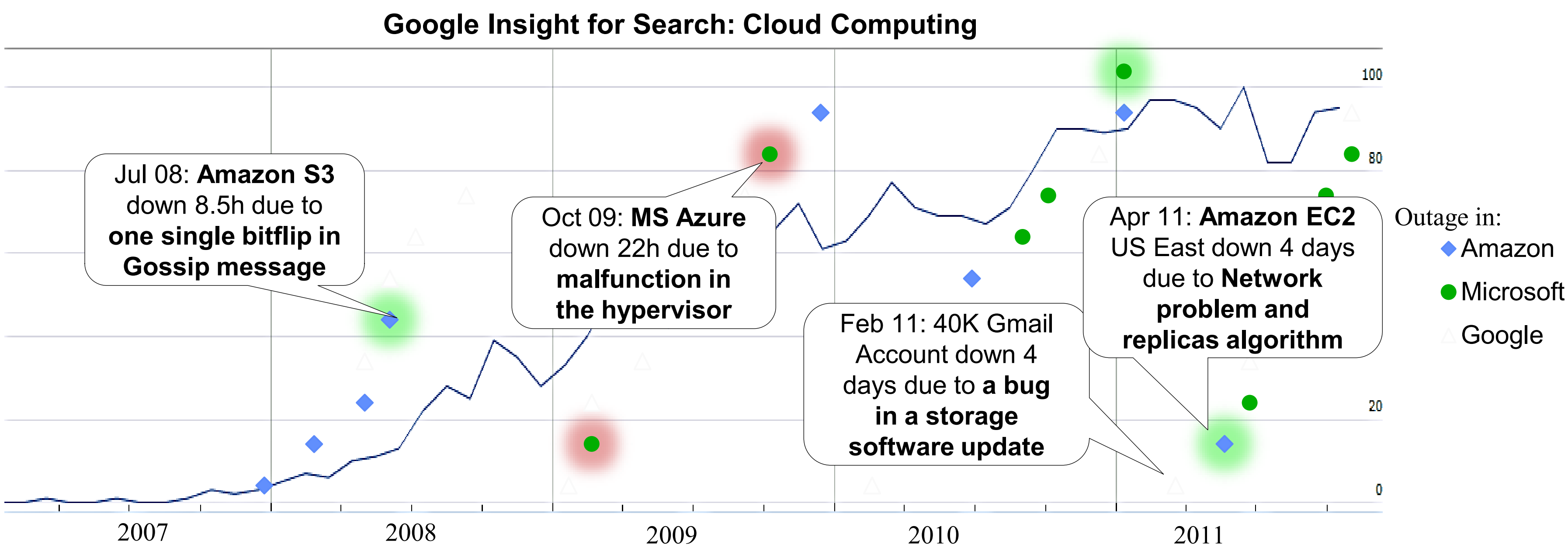

- Providing a higher level of reliability and availability is one of the biggest challenges of Cloud computing

Application to Reliability Evaluation Series and Parallel Systems

- Consider the problem of computing reliability of so-called series-parallel systems.
- A **series system** is one in which all components are so interrelated that the entire system will fail if any one of its components fails.
- A **parallel system** is one that will fail only if all its components fail.
- We will assume that failure events of components in a system are mutually independent. Consider a series system of n components.

Application to Reliability Evaluation (cont.)

- For I=1,2,…,n, define events A_i="Component *i* is functioning properly." The **reliability**, R_i, of component *i* is defined as the probability that the component is functioning properly. Then: $R_i=P(A_i)$.
- By the assumption of series connections, the system reliability:

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- If the system complexity is increased so that it contains ten similar components, its reliability is reduced 0.970^{10}=0.738.
- Imagine what happens to system reliability when a large system such as a computer system consists of tens to hundreds of thousands of components.

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- One way to increase the reliability of a system is to use redundancy, i.e., to replicate components with small reliabilities. This is called **parallel redundancy**.
- Consider a system consisting of n independent components in parallel; the system fails to function only if n components have failed.
- Define event A_i="The component i is functioning properly" and

  A_p="The parallel system of n components is functioning properly."

  Also let $R_i=P(A_i)$ and $R_p=P(A_p)$.

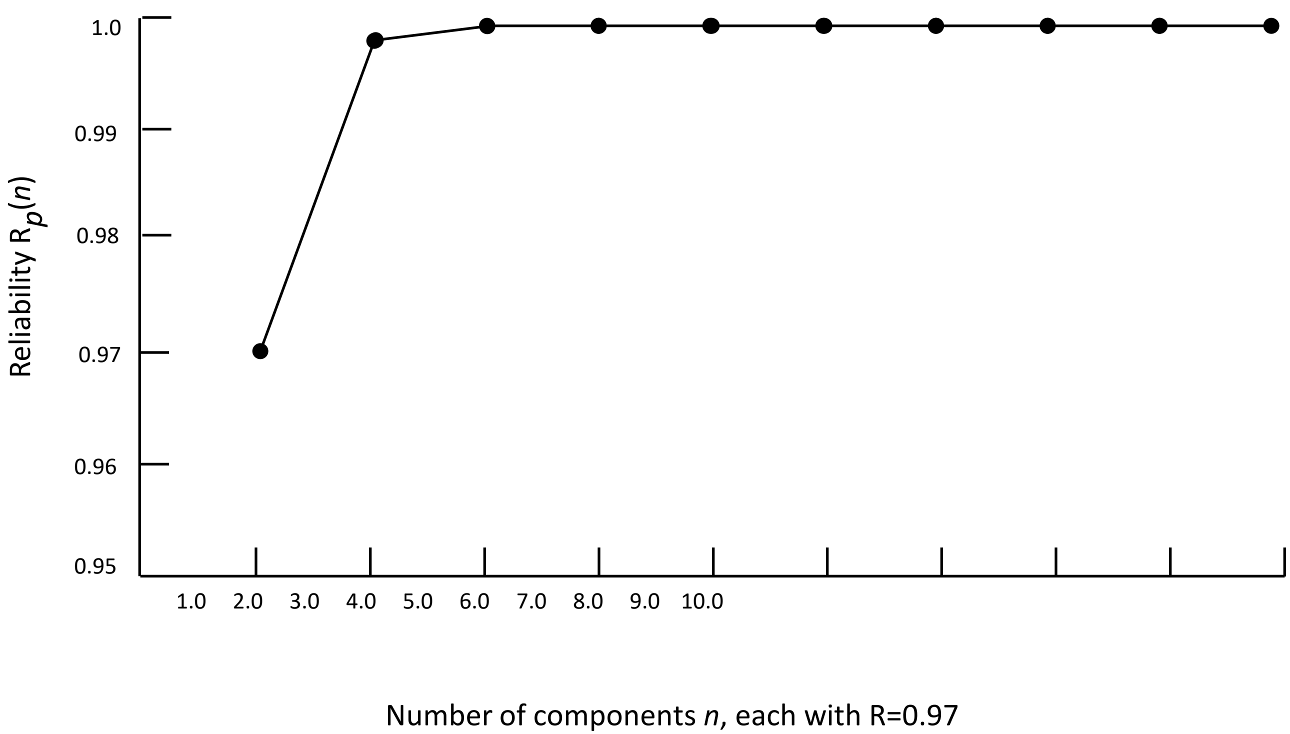
Product Law of Unreliabilities

- Thus, for parallel systems of *n* independent components, we have a **product law of unreliabilities** analogous to the **product law of reliabilities** of series systems.
- If we have a parallel system of five components, each with a reliability of 0.970, then the system reliability is increased to:

$$1-(1-0.970)^5 = 1-(0.03)^5 = 1-0.0000000243 = 0.9999999757$$

- However, we should be aware of the **law of diminishing returns**.

Reliability Curve of a Parallel Redundant System



Reliability of Series-Parallel Systems

- We can use formulas parallel and series systems in combination to compute the reliability of a system having both series and parallel parts (**series-parallel systems**).

- Consider a series-parallel system of n serial stages, where stage *i* consists of n_i identical components in parallel.

Reliability of Series-Parallel Systems (cont.)

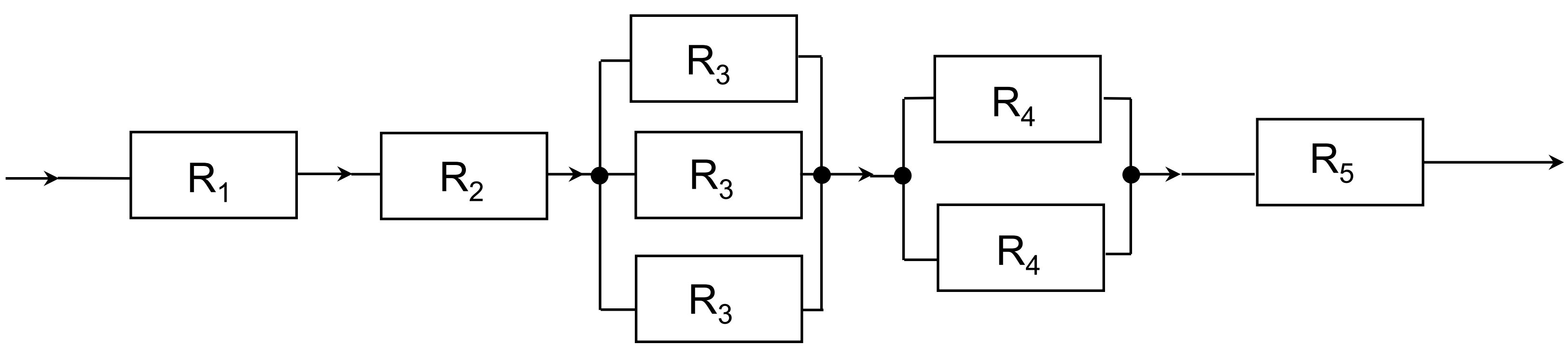
- The reliability of each component of stage *i* is R_i.
- Assuming that all components are independent, R_{sp} :

$$R_{sp} = \prod_{i=1}^{n} [1 - (1 - R_i)^{n_i}]$$

Series-Parallel System Example

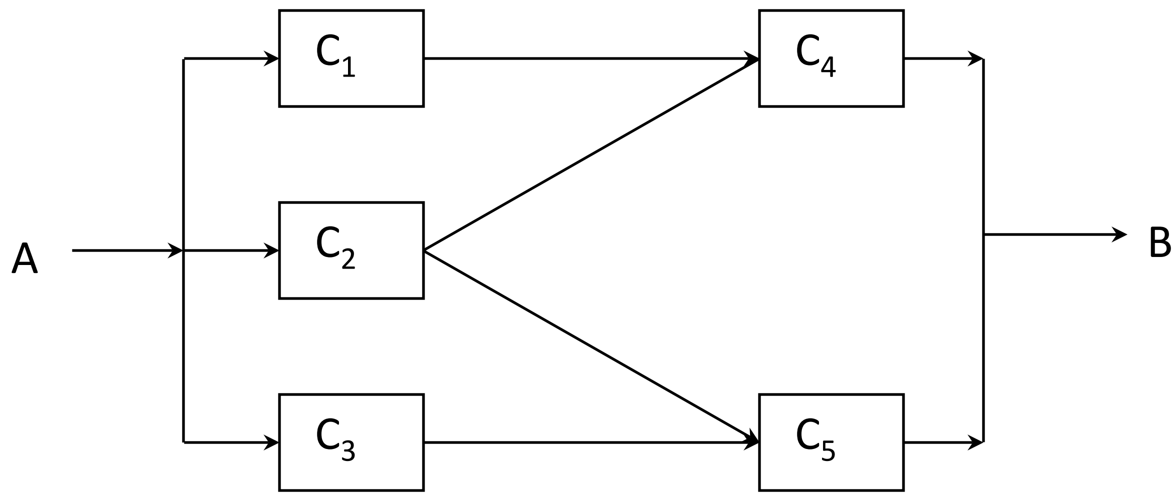
- Consider the series-parallel system shown in the diagram below, consisting of five stages, with $n_1=n_2=n_5=1$, $n_3=3$, $n_4=2$, and $R_1=0.95$, $R_2=0.99$, $R_3=0.70$, $R_4=0.75$, and $R_5=0.9$.

- Then: $R_{sp} = 0.95 \cdot 0.99 \cdot (1-0.3^3) \cdot (1-0.25^2) \cdot 0.9 = 0.772$



Application of Bayes
A More Complex System: Example

- Consider evaluating the reliability R of the five-component system. The system is said to be functioning properly only if all the components on at least one path from point A to point B are functioning properly.



Solution

In-Class Activity #1

- Divide into groups of 3 to 4 people
- Discuss / work together
- **Submit your own solution** (by 12:00)